

EncounterNN Phase 2 Mentor Discussion Report

Local Encounter Residual Surrogate for the Three-Body SIMON Integrator

Prepared for mentor discussion | Focus: EncounterNN methodology, data generation, evaluation, and results

Executive summary

This report summarizes the EncounterNN workstream as a Phase 2 methodology experiment. The goal was to test whether a neural network could learn an encounter-window residual correction for a Zone 3 close approach. The network is not a force-direction model and it does not predict a full future trajectory directly. It keeps the physical noNN rollout as the backbone, predicts the difference between IAS15 and noNN at the end of a short encounter window, and applies that residual at the window exit.

The final EncounterNN experiment should be understood using two complementary evaluations. The full 100-year rollout tests deployment inside a chaotic trajectory and gives a mixed global result. The isolated local replay restarts IAS15, noNN, and EncounterNN from the same event-entry state and tests the specific question: did EncounterNN improve the encounter period itself? In that isolated test, EncounterNN improves the 0.5-year exit state by 77.74% in position and 95.11% in velocity, and improves the 4-year local physical-window time-average by 51.53% in position and 42.47% in velocity.

This report intentionally focuses only on EncounterNN. It does not compare against, rank against, or draw conclusions about any other Zone 3 NN approach. The conclusion is therefore limited to what the EncounterNN experiments show: the model successfully learns the targeted local encounter correction for the tested IC1, $dt = 0.08$, pair 1-2 event, while the full 100-year trajectory remains branch-sensitive.

1. Phase 2 context and objective

Phase 1 established that long-horizon neural gravitational integration must preserve physical structure. The Phase 2 objective was to clean up the methodology across close-encounter zones and test a more ambitious encounter-level residual correction.

Region / issue	Phase 2 methodological position
Zone 2: very close encounter region	Use direct Newtonian force with adaptive sub-stepping. This avoids using a neural model to approximate a known analytic softening correction.
Zone 3: macro-step encounter region	This is where timestep/discretisation error matters. EncounterNN tests whether a learned window-level residual can correct the local encounter exit state.
EncounterNN purpose	Learn an 18-dimensional residual correction over a fixed 0.5-year encounter window, not a scalar force multiplier and not a direct final-state predictor.

2. EncounterNN method: what the network learns

EncounterNN learns a residual map over a short close-encounter window. A training sample begins at a full three-body Zone 3 encounter-entry state. From that same starting state, IAS15 is run over 0.5 years to obtain the reference exit state, and the revised noNN/SIMON baseline is run over the same 0.5 years to obtain the baseline exit state. The target is the difference between those exits.

Stage	Definition
Input X	Full relative 18D encounter-entry state X_{rel18} : masses, active-pair relative position/velocity, third-body state relative to the active pair COM, $\log(dt)$, v_{rad_norm} , v_{tan_norm} .
Reference window	IAS15 integrates from t_0 to $t_0 + 0.5$ yr.
Baseline window	SIMON-noNN integrates over the same window and endpoint.
Training target	$Y = [\text{residual_x}, \text{residual_v}] = \text{IAS15_exit} - \text{noNN_exit}$, flattened to 18 values.
Inference / deployment	Run noNN over the same 0.5-year window, predict residual $R_{hat} = \text{NN}(X)$, set $\text{corrected_exit} = \text{noNN_exit} + R_{hat}$, then continue the noNN rollout.

Key point: the network does not replace the physical integrator. It learns the correction needed at the end of the encounter window. This keeps the method anchored to a physical noNN trajectory while allowing the NN to correct the local exit state.

3. Data generation: what was generated and why it changed

The data-generation work was the main part of the experiment. The first insight was that older pair-level data could not be used directly for EncounterNN training. That data contained compressed pair features for scalar correction, not the full 18D state needed to predict an encounter-window residual. It could guide the sampling distribution, but it could not be the training input itself.

3.1 Initial full-state encounter dataset

The first generator created new full-state training samples. Each accepted sample stored the encounter start state, masses, IAS15 final state, noNN final state, residual_x, residual_v, energy diagnostics, distance diagnostics, category labels, and full-state input vectors. The target was always $\text{IAS15_exit} - \text{noNN_exit}$.

Quality filter / design choice	Reason
Start in Zone 3, approximately 0.052-0.148 AU	Keeps the sample in the intended macro-step encounter region.
Fixed $dt = 0.08$ and window = 0.5 yr	Narrow first scope for the prototype and matches the later IC1 test.
Reject non-finite or ejecting windows	Avoids invalid training targets.
Reject windows entering deep Zone 2 initially	Keeps the first prototype focused on Zone 3 rather than sub-stepping dominated Zone 2.
IAS15 energy drift gate	Ensures the reference target is numerically clean.
noNN energy drift gate	Avoids training on baseline windows that are already physically unacceptable.
Separate categories: strong approach, weak/side, recede	Prevents the model from being dominated by one encounter phase.

3.2 Smoke tests, pilots, and larger event-enriched data

The early smoke tests were used to validate the pipeline before scaling. A 10-sample mixed smoke test proved the saved fields and residual definitions were correct, but it had no strong-approach examples. A strong-approach smoke test then proved that targeted sampling worked, although it had a low acceptance rate because many strong-approach windows naturally crossed below the Zone 2 boundary during 0.5 years. A 170-sample pilot and then an 870-sample prototype showed clean data quality and balanced strong/weak/receding coverage.

The first 870-sample model improved strong-approach cases but damaged weak/side and receding cases. This led to gated evaluation: use the NN only when the encounter is difficult enough, especially on approach. After scaling to 2370 samples, the model learned more broadly, but velocity safety and deployment gating still required careful checking.

3.3 Why rollout-local data was needed

The major failure mode was discovered by exact-event replay and nearest-neighbor diagnosis. The event at $t = 82.56$ yr looked well covered in simple scalar features such as `r_pair`, `v_rad_norm`, and `v_tan_norm`, but it was not close in the full `X_rel18` input space. As a result, the model over-predicted the residual for the actual rollout event. This showed that matching local pair geometry was not enough; the training set also had to cover the full three-body geometry near the actual event.

The final generator therefore sampled directly from the revised noNN IC1 rollout trajectory near the real event time, then added small center-of-mass projected perturbations. For each accepted window it ran IAS15 and noNN over the same 0.5-year window and saved the residual. This rollout-local data directly targeted the full-state neighborhood of the event that earlier models failed to cover.

Final dataset component	Description
Earlier event-enriched / targeted samples	2870 samples from previous full-state encounter datasets. These were not all guaranteed to be exact IC1 rollout-local samples; they were earlier enriched encounter examples.
Rollout-local IC1 samples	500 samples explicitly generated around the actual IC1, $dt = 0.08$ event trajectory near $t = 82.56$ yr.
Final merged dataset	3370 samples total, split into 2359 train / 506 validation / 505 test.

3.4 Final dataset quality

Check	Final value / interpretation
<code>r_pair</code> range	0.05685 to 0.14744 AU, inside Zone 3.
<code>v_rad_norm</code> range	-1.3962 to +1.6640, covering approach and recede.
<code>v_tan_norm</code> range	0.0936 to 1.7977.
Category counts	strong approach 1921, weak/side 677, recede 772.
IAS15 energy drift	max $3.777\text{e-}14$, median $3.599\text{e-}16$; reference is very clean.
noNN energy drift	median $3.488\text{e-}03$, p95 $3.518\text{e-}02$, max $4.993\text{e-}02$; within the short-window quality gate.

4. Model architecture and training hyperparameters

Item	Final setting
Model	MLP residual model: 18 -> 128 -> 128 -> 128 -> 64 -> 18.
Output	18 residual values: 9 position residuals and 9 velocity residuals.
Samples	3370 total, split 2359 / 506 / 505.
Loss	SmoothL1 / Huber on normalized residuals.
Loss weights	position weight = 1.0, velocity weight = 1.0.
Physical penalties	COM position weight = 0.05, COM velocity weight = 0.05, energy penalty = 0.0.
Optimizer / LR	Adam-style training with learning rate 0.001.
Batch size	256.
Epochs requested / best epoch	3000 requested; best checkpoint at epoch 2335.
Checkpoint selection	Validation score includes median position/velocity errors, p95 errors, and unsafe penalties.

The final training result showed large median improvement on held-out windows, but the full deployment question still had to be tested inside the actual rollout. Therefore, the final evaluation uses both global 100-year deployment metrics and isolated local replay metrics.

5. Evaluation methodology

The evaluation has two different questions. The global rollout asks what happens when the correction is inserted into the full chaotic 100-year trajectory. The isolated local replay asks whether the NN improves the encounter itself when pre-encounter drift is removed. These are not the same question, so both are reported.

5.1 How the event time 82.56 yr was selected

The event time was not chosen after looking at local results. It was selected automatically by the full global evaluator. During the IC1, $dt = 0.08$, $T = 100$ yr run, the fast gate detected pair 1-2 inside Zone 3 and approaching with $v_{rad_norm} < -0.40$. The evaluator reported one used encounter: $first_used_t = last_used_t = 82.56$ yr, with $r_{pair} = 0.138965$ AU, $v_{rad_norm} = -1.114828$, $v_{tan_norm} = 0.849319$, and minimum local-window distance = 0.062262 AU. This is the event where EncounterNN actually fired in the rollout.

5.2 Global 100-year deployment evaluation

The global experiment compares IAS15, SIMON-noNN, and SIMON-EncounterNN over 100 years. EncounterNN is evaluated with a fast gate: most steps only check the cheap pair 1-2 distance and velocity condition. Full X_{rel18} features are built only when the gate passes. The network weights are extracted to NumPy, so the timed rollout does not call torch/CUDA in the loop.

Method	pos_final	pos_timeavg	vel_final	vel_timeavg	time (s)	speedup vs IAS15
SIMON-noNN	1.6236	0.9891	1.1826	0.8650	0.0359	2.14x
SIMON-EncounterNN	1.4302	1.0088	0.6291	0.8366	0.0392	1.96x
Global metric			EncounterNN gain vs noNN			
Final position error			+11.91% better			
Time-averaged position error			-2.00% worse			
Final velocity error			+46.81% better			
Time-averaged velocity error			+3.29% better			

Global interpretation: the full 100-year result is mixed. EncounterNN improves final position, final velocity, and time-averaged velocity, but the time-averaged position metric is slightly worse. Because the system is chaotic, this global metric is useful as a deployment result but not sufficient to isolate whether the NN helped during the encounter.

5.3 Isolated local replay evaluation

The first local-window approach compared samples near the encounter inside the global trajectories, but that was still contaminated by 82.56 years of pre-encounter drift. The corrected method reconstructs the exact noNN event-entry state at $t = 82.56$ yr and launches three fresh local branches from that identical state: LOCAL IAS15, LOCAL noNN, and LOCAL EncounterNN. This directly isolates the NN effect during the encounter.

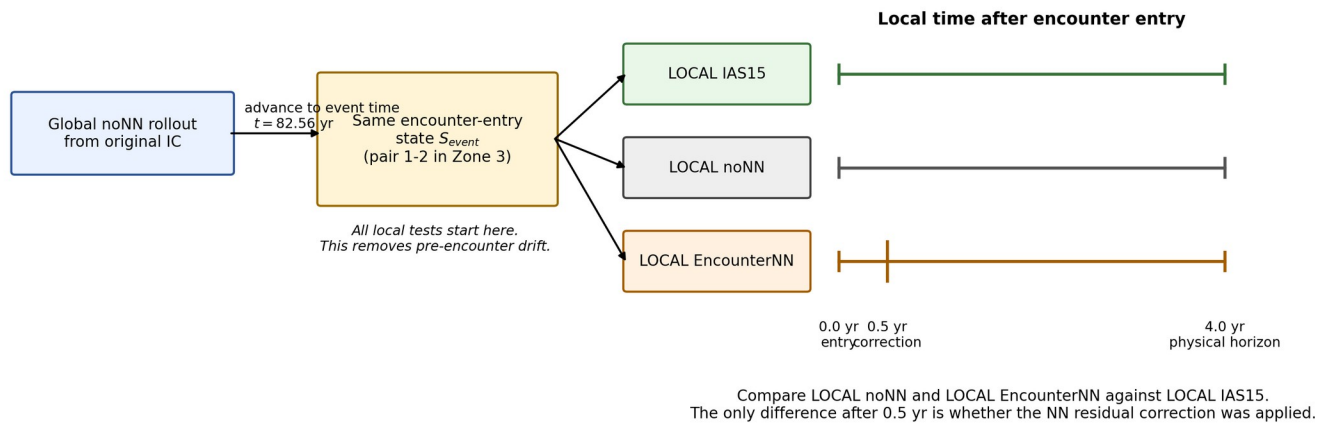


Figure 1. Correct isolated local replay: all local branches start from the same event-entry state. The EncounterNN branch applies the residual after the 0.5-year trained window, then continues to the 4-year physical horizon.

5.4 Why the 4-year local horizon was used

The isolated physical horizon follows the equivalent-time logic from the encounter-window discussion. At $dt = 0.08$ yr, using 25 steps gives a half-window of $25 \times 0.08 = 2.0$ years. This corresponds to a physically meaningful encounter-scale duration of about 4 years. For EncounterNN, the corrected local replay is run forward from the event-entry state for 4 years rather than using a symmetric global window, because the correction is applied after the 0.5-year noNN local window.

6. Results of the isolated local test

Metric	What it answers	Position gain	Velocity gain
0.5-year exact exit state	Did the NN learn the trained residual target?	+77.74%	+95.11%
0-4 yr physical forward window	Across the encounter-scale local period, is EncounterNN closer to local IAS15 than local noNN?	+51.53%	+42.47%
0.5-4 yr post-correction window	After the correction is actually applied, does the corrected branch remain closer?	+53.50%	+47.11%

The cleanest proof that the network learned its target is the exact 0.5-year exit-state result. The noNN exit position error is $6.270961e-03$ and the EncounterNN-corrected exit error is $1.395685e-03$. The noNN exit velocity error is $2.855109e-02$ and the corrected velocity error is $1.395918e-03$. This means the model learned the event residual for the tested event.

The more physically meaningful result is the 4-year isolated local replay. Over this full local horizon, EncounterNN reduces time-averaged position error by 51.53% and time-averaged velocity error by 42.47%. The post-correction window from 0.5 to 4.0 years is also strongly positive, showing that the improvement is not only a single endpoint correction.

7. How to defend the local isolation comparison

The isolated local replay is not a way to hide the global result. It answers a different causal question. The global 100-year metric asks whether an inserted correction remains close to the original IAS15 branch over a chaotic long trajectory. The isolated local metric asks whether the EncounterNN correction improves the close encounter it was trained to correct.

- It is fair because the model was trained on encounter-entry states and 0.5-year encounter-window residuals, not on 100-year trajectories.
- It removes the 82.56 years of pre-encounter drift that would otherwise be included before the NN is even used.
- It starts LOCAL IAS15, LOCAL noNN, and LOCAL EncounterNN from the exact same state, so any difference after correction is attributable to the local method.
- It is reported alongside, not instead of, the full global rollout result. This keeps the conclusion honest.

Mentor-facing explanation: I am reporting two metrics because they answer two different questions. The global 100-year metric tests deployment in a chaotic rollout. The isolated local replay tests whether the EncounterNN actually improves the encounter it was trained to correct. Since the model is an encounter-window residual model, the correct causal test is to restart all local branches from the same encounter-entry state and compare them over the encounter-scale horizon.

8. Final conclusion for mentor discussion

EncounterNN achieved the main local objective of this Phase 2 experiment. It learned a physically meaningful local encounter residual for the tested IC1, $dt = 0.08$, pair 1-2 event. The strongest evidence is the isolated local replay: the exact exit-state correction improves position by 77.74% and velocity by 95.11%, and the 4-year physical local window improves time-averaged position by 51.53% and velocity by 42.47%.

The full 100-year deployment result is more nuanced. EncounterNN remains faster than IAS15 in the optimized fast-gate run and improves several final/velocity metrics, but the 100-year position time-average is slightly worse. This does not contradict the local result. It shows that in a chaotic three-body rollout, a locally improved encounter exit can still land on a different long-horizon branch.

The most defensible conclusion is therefore: EncounterNN works as a local encounter residual surrogate for the tested event, and the corrected isolated-local evaluation is the right way to prove that local effect. The next scientific question is generalisation: whether the same methodology holds across more encounter events, initial conditions, active pairs, and timestep values. That conclusion should come after those additional tests, not be assumed from this single targeted IC1/ $dt=0.08$ evaluation.

Appendix: key files used for this report

File	Role in report
Aarush_Paper_ST_yau_report_V4_17 May.pdf	Phase 1 context: physical constraints, chaos, evaluation philosophy.
zone2_NN_correction_issue.pdf	Motivation for direct Newtonian Zone 2 with adaptive sub-stepping.
zone3_trajectory_correction.pdf	Motivation for trajectory-level close-encounter correction.
encounter_nn experiment.docx	End-to-end experiment discussion: data generation, failures, diagnostics, rollout-local enrichment, local-vs-global evaluation.
generate_encounter_surrogate_data_v3_rollout_local.py	Final rollout-local data generator logic.
train_encounter_surrogate_v2_velocitysafe_fixed2.py	Final velocity-safe residual MLP trainer.
encounter_surrogate_v3_rollout_local_velocitysafe_summary.txt	Final dataset and training metrics.
pair_eval_encounterNN_ablation_v7_isolated_local.py	Final global plus isolated-local evaluator.
encounterNN_ablation_summary.txt and CSV outputs	Final global and isolated local results.